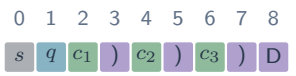
$$E = H[P_{opt}, :],\ D = H[P_{decide}, :], \quad k = EW_k^\mathsf{T} + b_k,\ q = DW_q^\mathsf{T} + b_q, \quad s = kq^\mathsf{T}/\sqrt{d_p}$$

1. Token layout

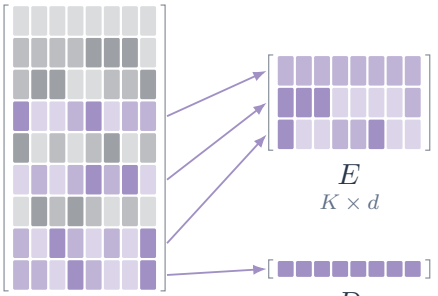


x
T integer token IDs

2. Kev

32 hybrid causal blocks
Qwen3.5-4B-Base;
LoRA r=16
one row per question

3. Contextual states and indices



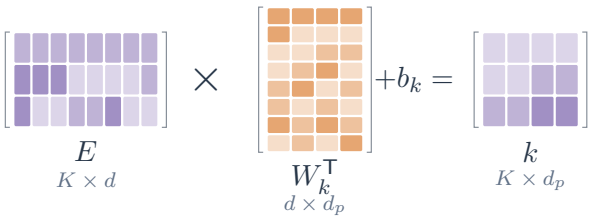
H
$T \times d$

E
$K \times d$

D
$1 \times d$

E: option-closing positions
D: decision marker
decision sees every option

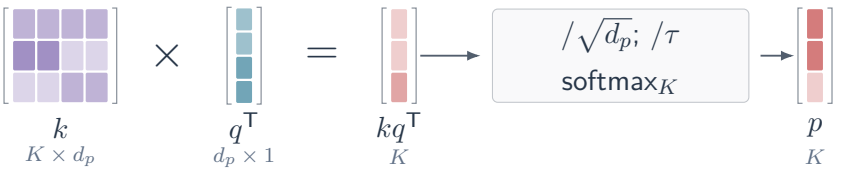
4. Separate biased projections



E $K \times d$ × W_k^T $d \times d_p$ $+b_k =$ k $K \times d_p$

D $1 \times d$ × W_q^T $d \times d_p$ $+b_q =$ q $1 \times d_p$

5. Query-option dot product



k $K \times d_p$ × q^T $d_p \times 1$ = kq^T K → $/\sqrt{d_p}$; $/\tau$ softmax$_K$ → p K

Axes	$d = 2560,\ d_p = 256$; Single-question illustration; batch omitted. Tile counts encode axes, not literal sizes.
Objects	Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.
Mechanism	Hybrid Kev-4B uses question rows, not the packed mask path. Option ends are causal; the decision query sees all options. Actual head temperature was not read. Pinned checkpoint configuration selects candidate CE. Two biased projections d-to-256; pointer scaling by sqrt(256) is distinct from fitted temperature.